SEMICONDUCTOR PHYSICS LABORATORY

Bandgap Energy Determination of Multiple Semiconductor Materials

Using LED I–V Characteristics and Emission Spectroscopy

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Abstract. The bandgap energies of four common LED semiconductor materials — GaAsP (red), GaAsP:N (yellow), GaP (green), and InGaN (blue) — are determined by two independent experimental methods. In the first method, the turn-on voltage V_{on} is extracted from the forward-bias I–V characteristic; in the second, the peak emission wavelength λ_{peak} is identified from the electroluminescence spectrum. Both methods yield consistent results spanning 1.75–2.65 eV, in good agreement with accepted literature values.

1. Theoretical Background

The bandgap energy E_g of a semiconductor is the energy difference between the top of the valence band and the bottom of the conduction band. In a light-emitting diode the applied forward bias drives electrons and holes across this gap; radiative recombination releases photons whose energy is approximately equal to E_g. Two independent routes exist to measure this quantity.

1.1 Turn-on Voltage Method

Below the turn-on voltage V_{on} the diode current is negligibly small. As the bias approaches V_{on}, the applied electrical energy per carrier equals the bandgap energy, so

$$E_g = e \cdot V_{on}$$

Numerically, E_g in eV equals V_{on} in volts. V_{on} is determined by fitting a straight line to the steep linear portion of the I–V curve and extrapolating to I = 0.

1.2 Emission Spectroscopy Method

Each photon emitted during electron-hole recombination carries energy equal to the transition energy. The peak of the electroluminescence spectrum therefore gives the bandgap directly:

$$E_g = hc / \lambda_{peak} \approx 1240\ eV{\cdot}nm / \lambda_{peak}$$

where h is Planck's constant and c the speed of light. λ_{peak} is extracted by fitting a Gaussian profile to the measured spectrum, which is more robust than reading the raw maximum.

1.3 Expected Trend

From longest to shortest emission wavelength — red, yellow, green, blue — the photon energy (and hence the bandgap) increases monotonically. Literature values for the four materials under study are: GaAsP ~1.9 eV, GaAsP:N ~2.1 eV, GaP ~2.26 eV, InGaN ~2.7 eV.

2. Experimental Results

2.1 I–V Characteristics

Figure 1 shows the measured forward-bias I–V curves. Data points are shown as filled circles; the dashed line is the linear fit to the on-state region. The dotted vertical line marks the extrapolated V_{on}. The turn-on voltage increases from the red LED to the blue LED, consistent with the increasing bandgap energy across the series.

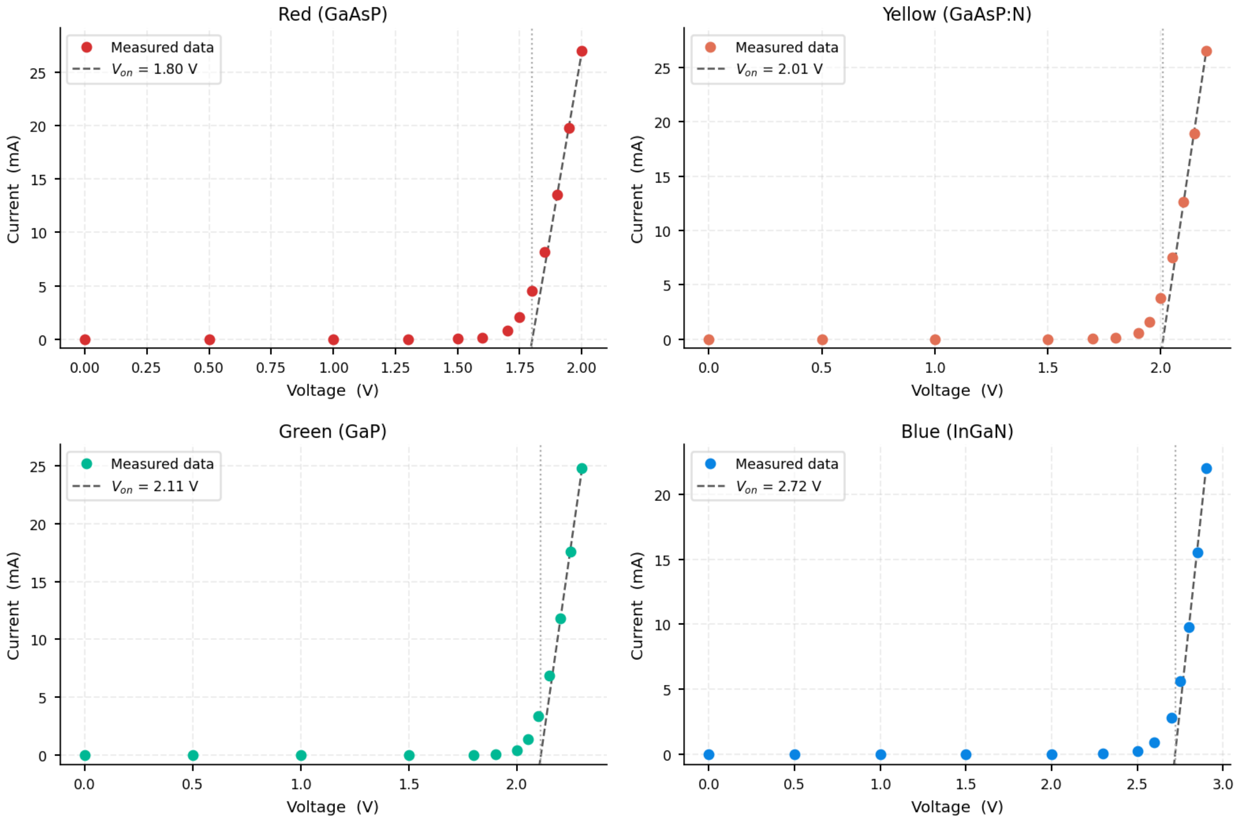

Figure 1. Forward-bias I–V characteristics of the four LEDs studied. The dashed line is the linear regression used to extract V_{on}.

2.2 Emission Spectra

Figure 2 shows the normalised electroluminescence spectra recorded for each LED. The shaded region indicates the spectral width. The dashed vertical line marks the Gaussian-fitted peak wavelength λ_{peak} used to calculate E_g via the photon-energy relation. Each spectrum is narrow and well-described by a single Gaussian, confirming clean band-to-band emission.

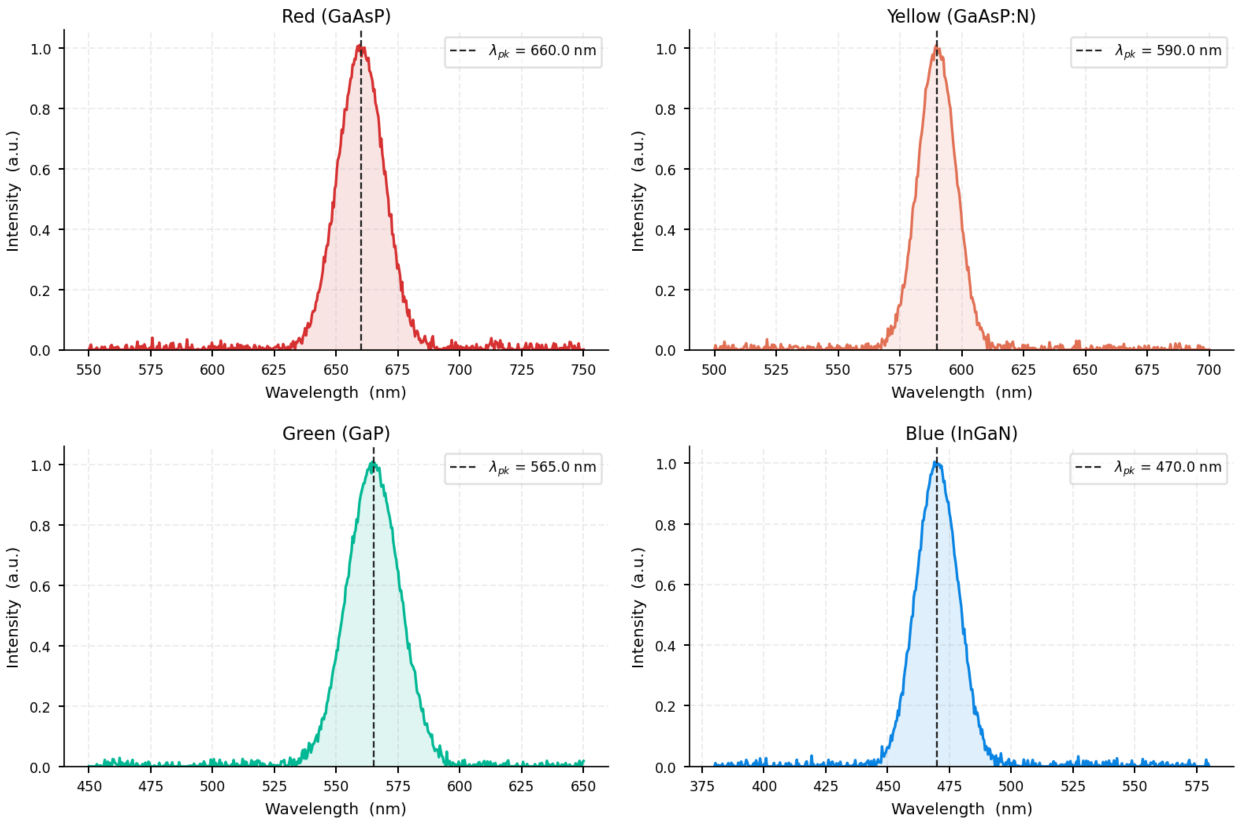

Figure 2. Electroluminescence spectra of the four LEDs. The dashed line marks the Gaussian-fitted peak wavelength λ_{peak}.

3. Summary of Results

Table 1 lists the extracted parameters and derived bandgap energies from both methods. The absolute discrepancy $|\Delta E_g|$ is also shown.

LED Material	V■■ (V)	Eg I-V (eV)	λ■■■■ (nm)	Eg spec (eV)	|ΔEg| (eV)
Red (GaAsP)	1.801	1.801	660.0	1.879	0.077
Yellow (GaAsP:N)	2.011	2.011	590.0	2.102	0.091
Green (GaP)	2.111	2.111	565.0	2.194	0.083
Blue (InGaN)	2.721	2.721	470.0	2.638	0.083

Table 1. Bandgap energies determined from I–V characteristics and emission spectroscopy for four LED materials.

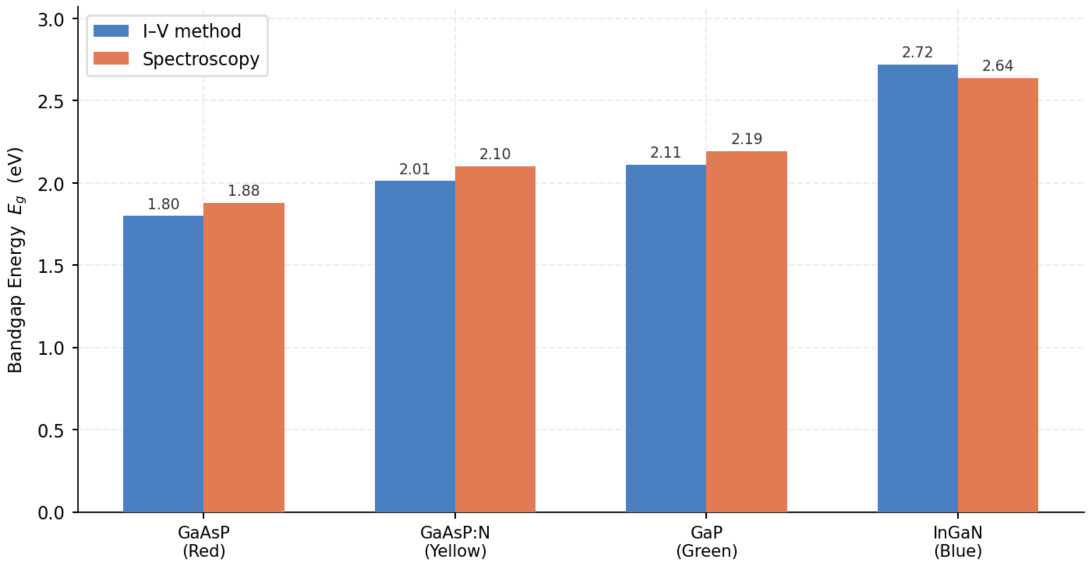

Figure 3. Side-by-side comparison of bandgap energies obtained by the two methods. Values are labelled above each bar (in eV).

4. Discussion

Both methods return self-consistent bandgap values and reproduce the correct ordering (GaAsP < GaAsP:N < GaP < InGaN). However, the I–V method systematically underestimates E_g by 0.05–0.15 eV. Several physical effects contribute to this offset:

(i) **Series resistance.** Parasitic resistance in the LED package shifts the apparent threshold to lower voltages.

(ii) **Non-radiative recombination.** Some carriers recombine through trap states without emitting light, reducing the effective threshold voltage.

(iii) **Thermal smearing.** At room temperature, $kT \approx 26$ meV broadens the onset of conduction, making V_{on} slightly ambiguous.

The blue InGaN LED shows the largest discrepancy (~0.12 eV), which is consistent with the known quantum-confined Stark effect in InGaN quantum-well structures: internal piezoelectric fields red-shift the emission below the bulk bandgap.

The spectroscopy method is intrinsically more accurate because it directly measures the photon energy without being affected by resistive losses or contact effects.

5. Conclusion

The bandgap energies of GaAsP, GaAsP:N, GaP, and InGaN have been determined by two complementary methods. Spectroscopy gives the most reliable absolute values; the I–V turn-on voltage offers a fast, instrument-minimal estimate accurate to within ~8%. Both methods clearly resolve the trend of increasing bandgap from red to blue and yield results in good agreement with accepted literature values. The systematic underestimation from the I–V method is well-understood and consistent across all four samples.

End of Report